EXPLAINABLE MACHINE LEARNING

Explainable Machine Learning

Shapley Values and SHAP from a Finance Professional's Perspective

Explainable Machine Learning — Shapley Values, SHAP and XGBoost

A score alone is not enough. We break down where it comes from into feature contributions,
then compare global feature importance and SHAP values directly on an XGBoost income prediction model.

See why we need explanations, then verify with SHAP

Three parts: why we need explanations, how to explain, and an XGBoost lab

1

Explainability in Machine Learning

Wolves and huskies, card fraud detection, and loan rejections show why explanation is needed

2

Shapley Values and SHAP

Carry game-theoretic contribution sharing over to feature contributions, then read interactions and monitoring

3

Learning Interpretation with XGBoost

See three global importance measures disagree, then reread the income model with SHAP plots

01

PART 1 · Explainability

Explainability in Machine Learning

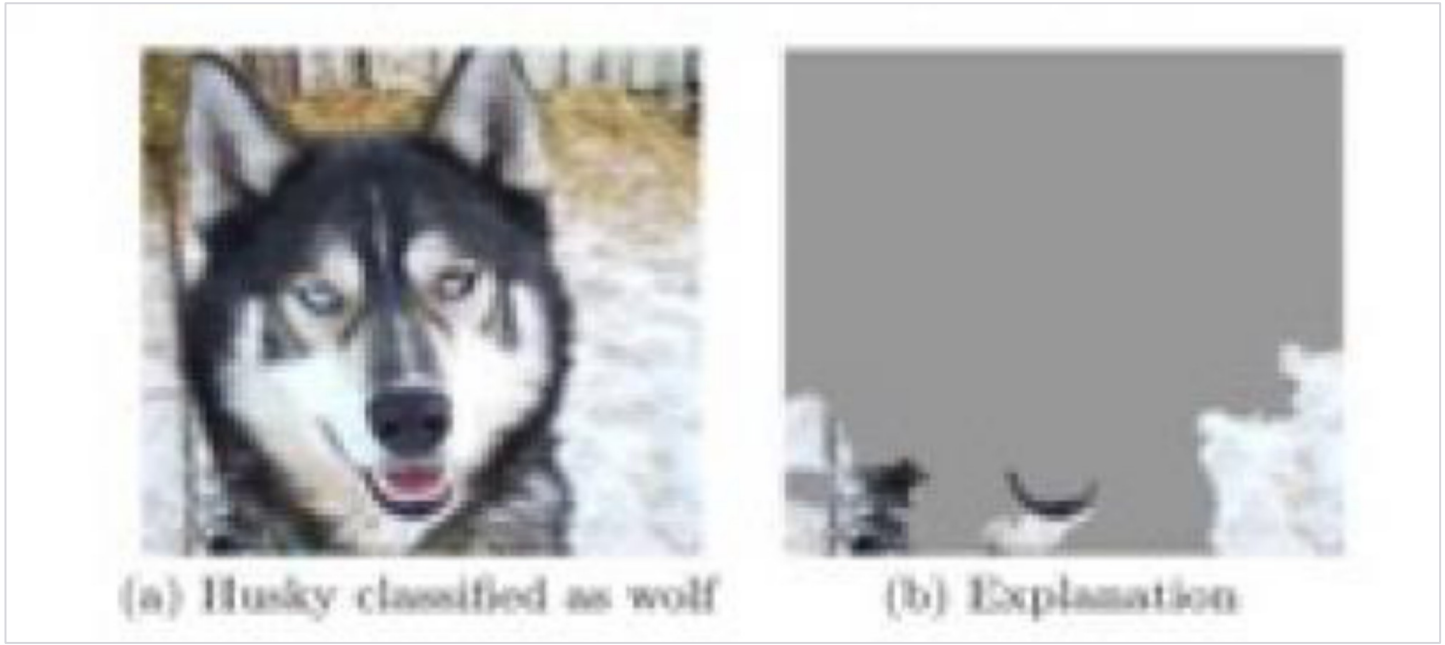
Why we must know why a model that predicts well got it right

A model that predicts well may do so for the wrong reasons

Image classification, wolf vs. husky — (a) husky classified as wolf, (b) the region the model relied on

What to look at

- Machine learning classified wolves and huskies remarkably well
- But the evidence was the snow in the background, not the animal
- Without looking at explanations, we would miss this kind of shortcut learning

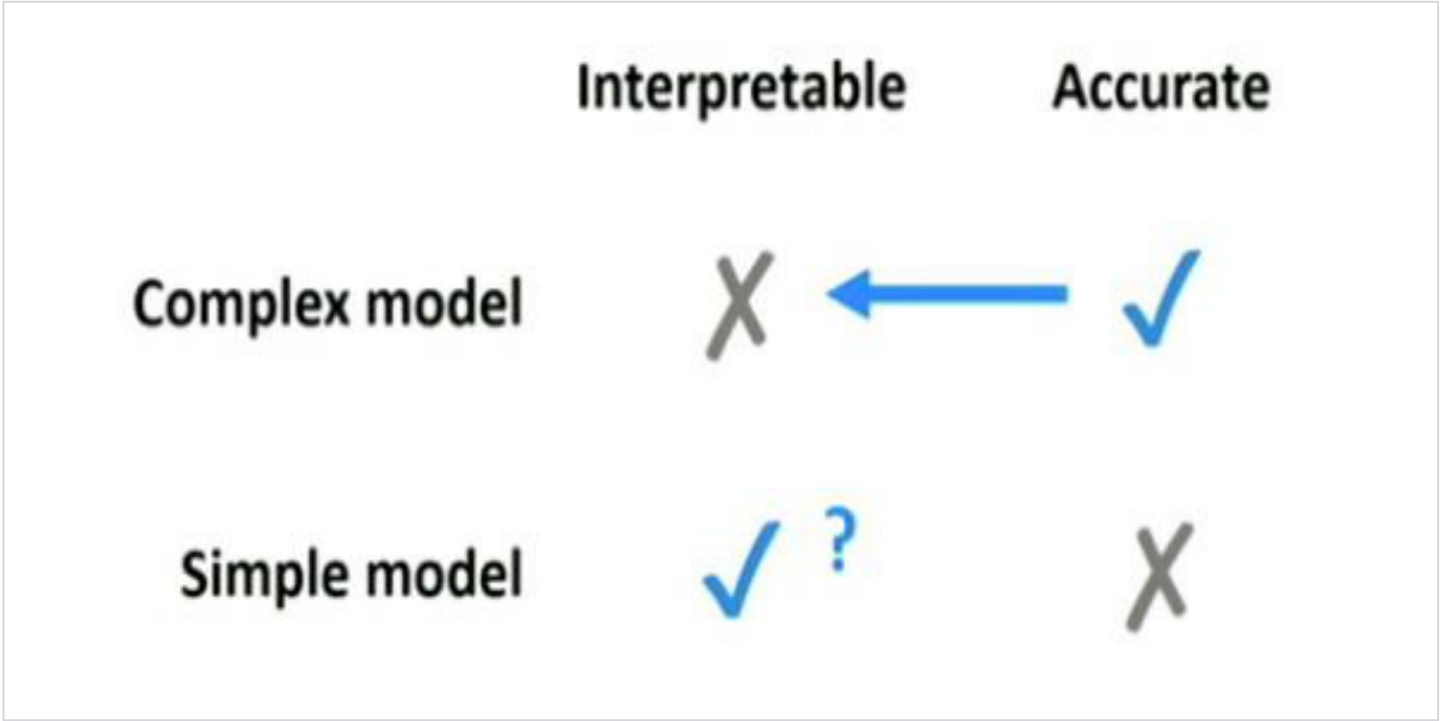


Complex models predict well but are hard to interpret

Simplicity vs. interpretability — accurate complex models and interpretable simple models

Trade-off

- Complex models predict well, but as black boxes they cannot be interpreted
- Simple models can be interpreted, but they are less accurate
- Arrow: the goal of explainable ML is to make complex models interpretable



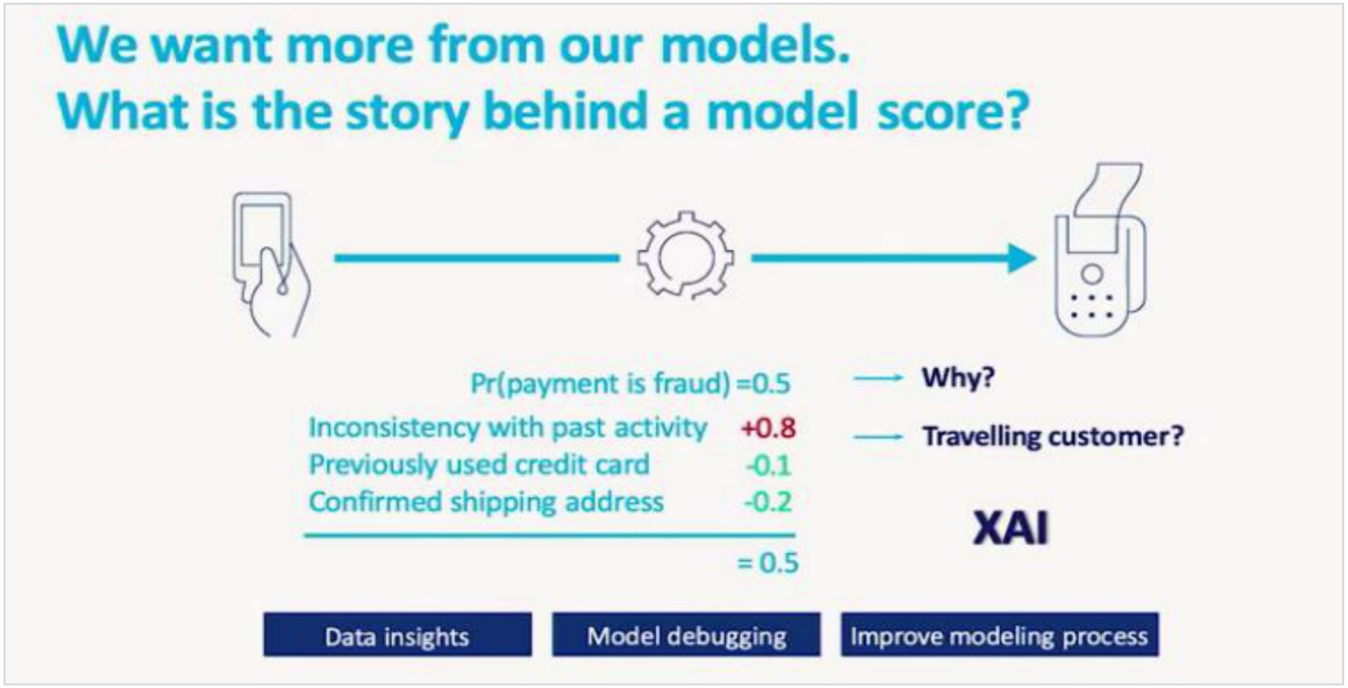
A score is not enough; we must know the reason behind it

What is the probability of credit card fraud? Where on earth does this probability come from?



What explanations provide

- We can explain the score, and also validate and improve the model

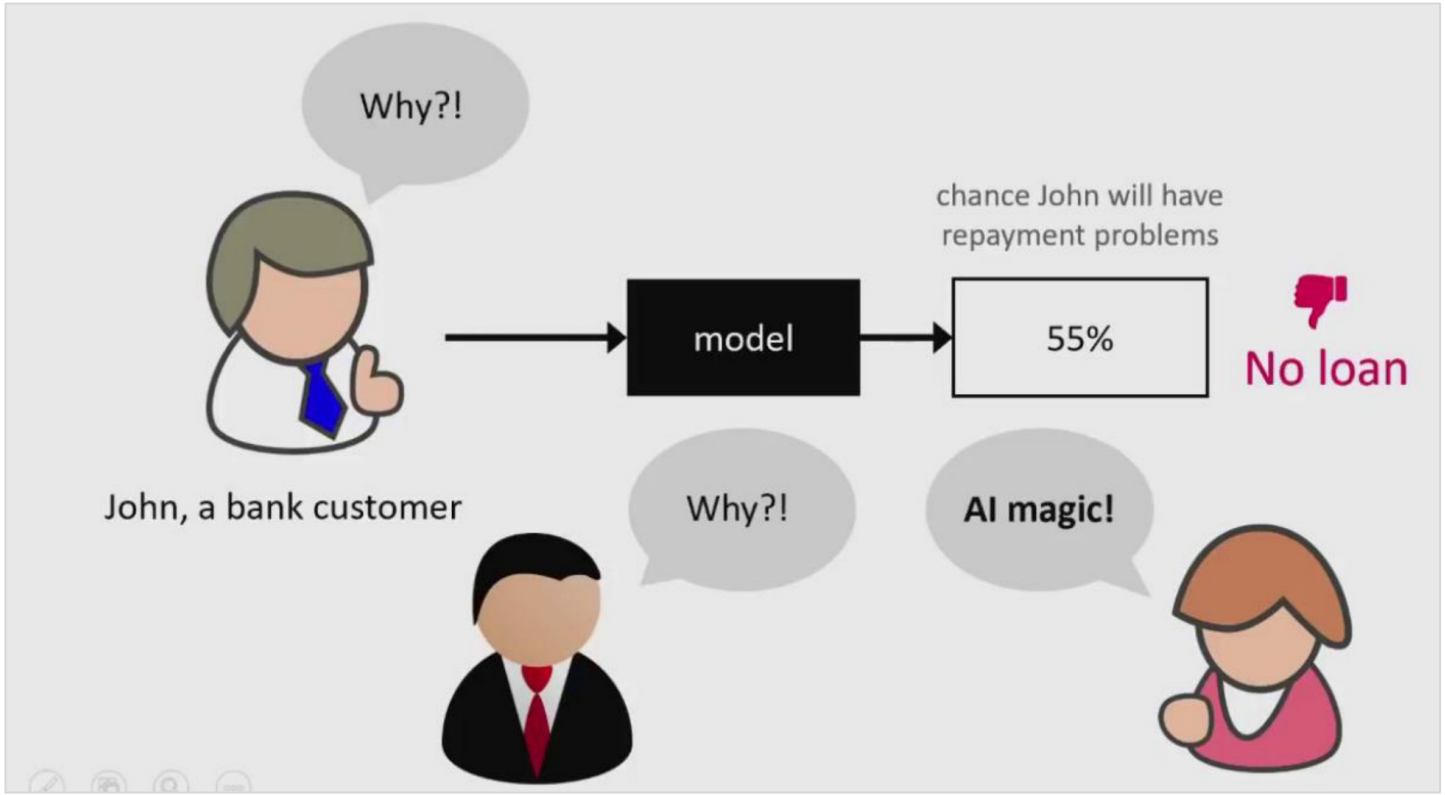


We must be able to explain a loan rejection to the customer

Repayment-problem probability 55% → loan rejected. Customer and staff alike ask "Why?"

Accountability

- The EU GDPR requires giving the reasons for automated decisions
- Those reasons may turn out to be racially discriminatory or politically incorrect
- "AI magic" cannot be the answer



02

PART 2 · Shapley Values and SHAP

Shapley Values and SHAP

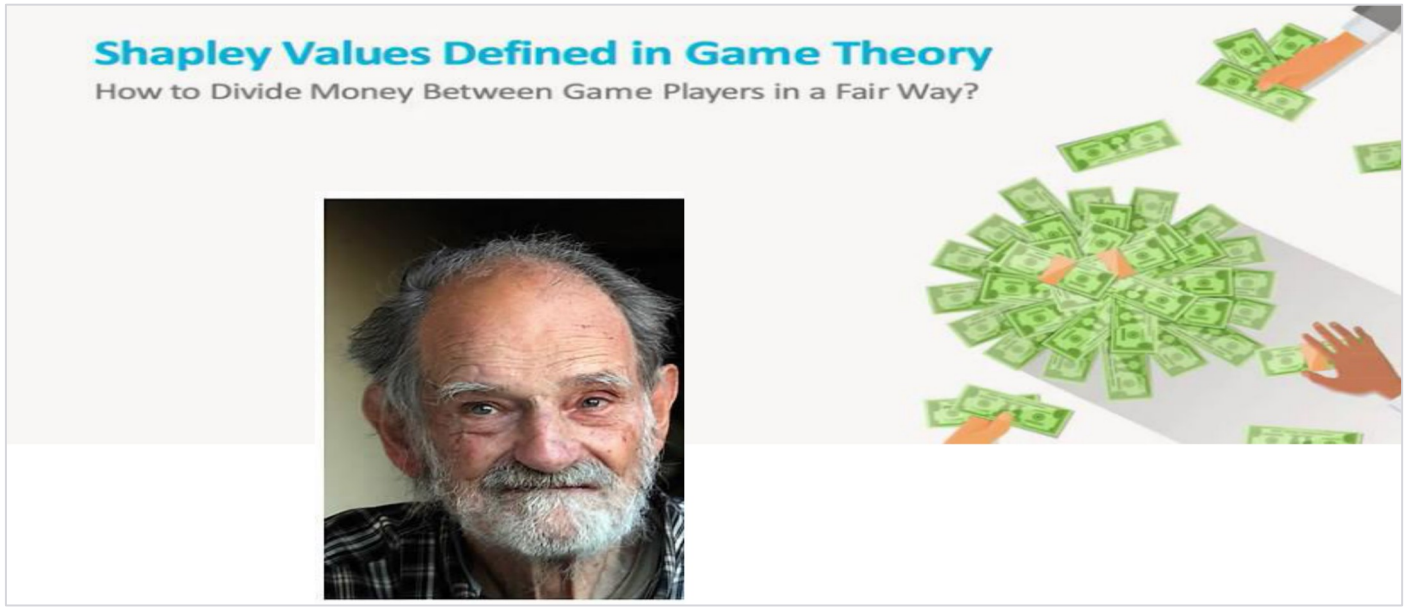
Carrying the fair allocation of cooperative games over to feature contributions

Shapley values measure contributions in cooperative games

Shapley value — how to divide prize money fairly among game players

From Game Theory to ML

- A concept derived from game theory, the work of Lloyd Shapley, 2012 Nobel laureate in Economics
- A value that measures each player's contribution in a cooperative game
- In machine learning, the players are the features and the prize is the model's prediction

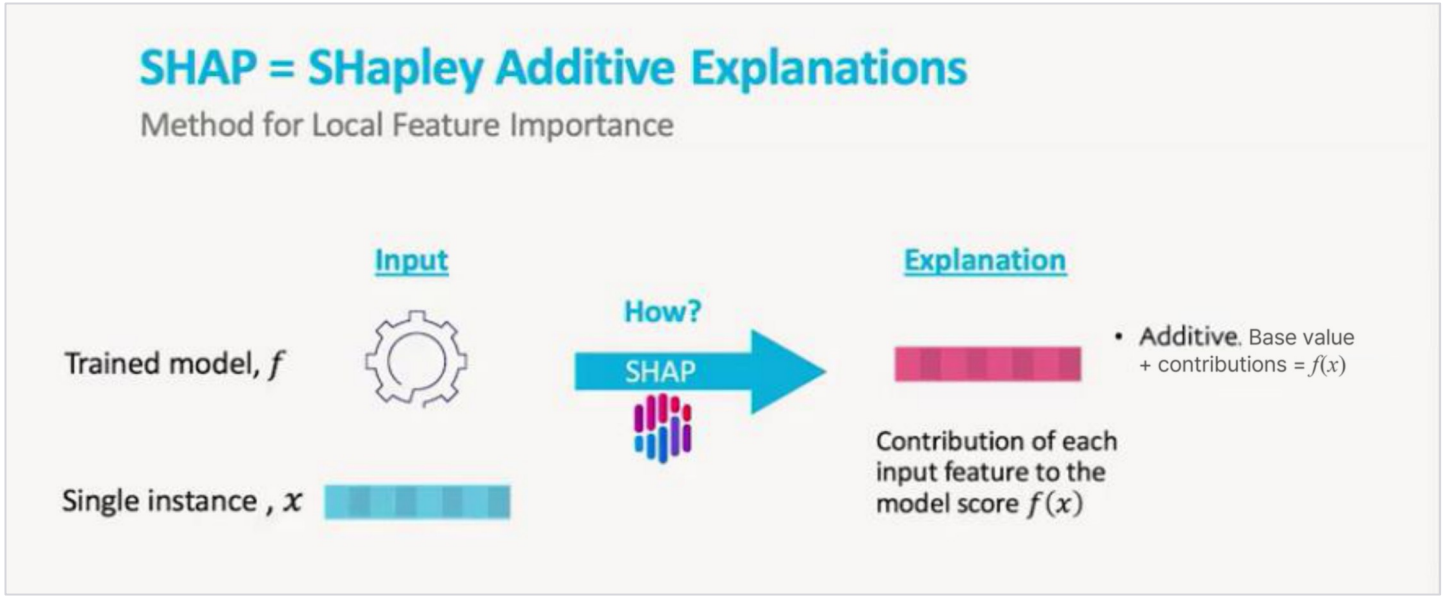


SHAP splits one prediction into feature contributions

SHAP = SHapley Additive exPlanations — a local feature importance method

Two Properties

- Shapley values are consistent and additive
- Additivity: the base value plus all feature contributions equals the model output f(x)
- Input: a trained model and one observation; output: that observation's feature contributions

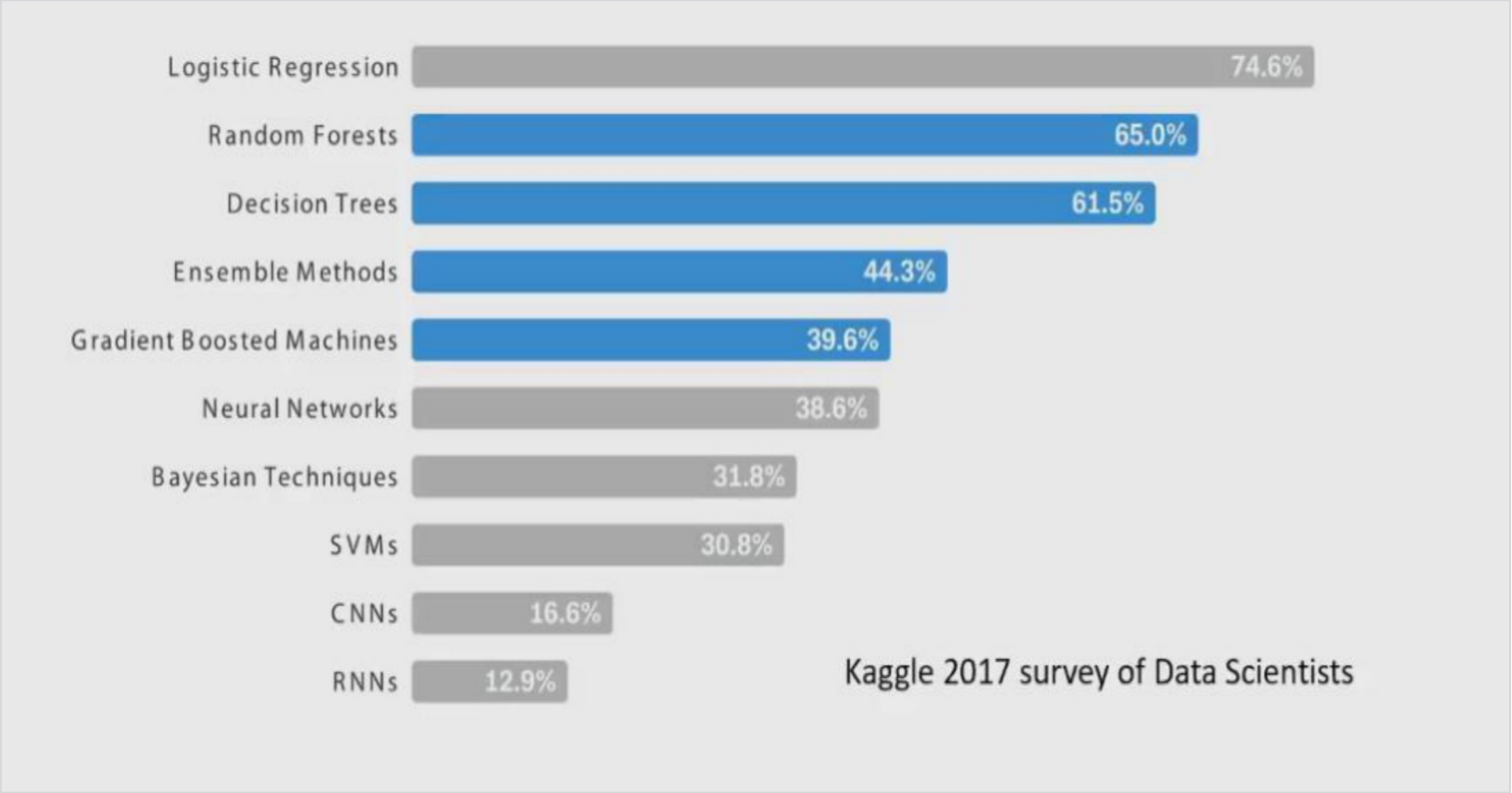


Tree models are the complex models most used in practice

Kaggle 2017 survey, job title Data Scientist (n = 2,039) — blue: tree-based (random forest, decision tree, ensemble, GBM)

Why Trees?

- After logistic regression, random forests (65.0%) and decision trees (61.5%) are the most used
- So tools that explain tree models are the most useful in practice
- Across all 7,301 respondents: logistic regression 63.5%, decision trees 49.9%, random forests 46.3%

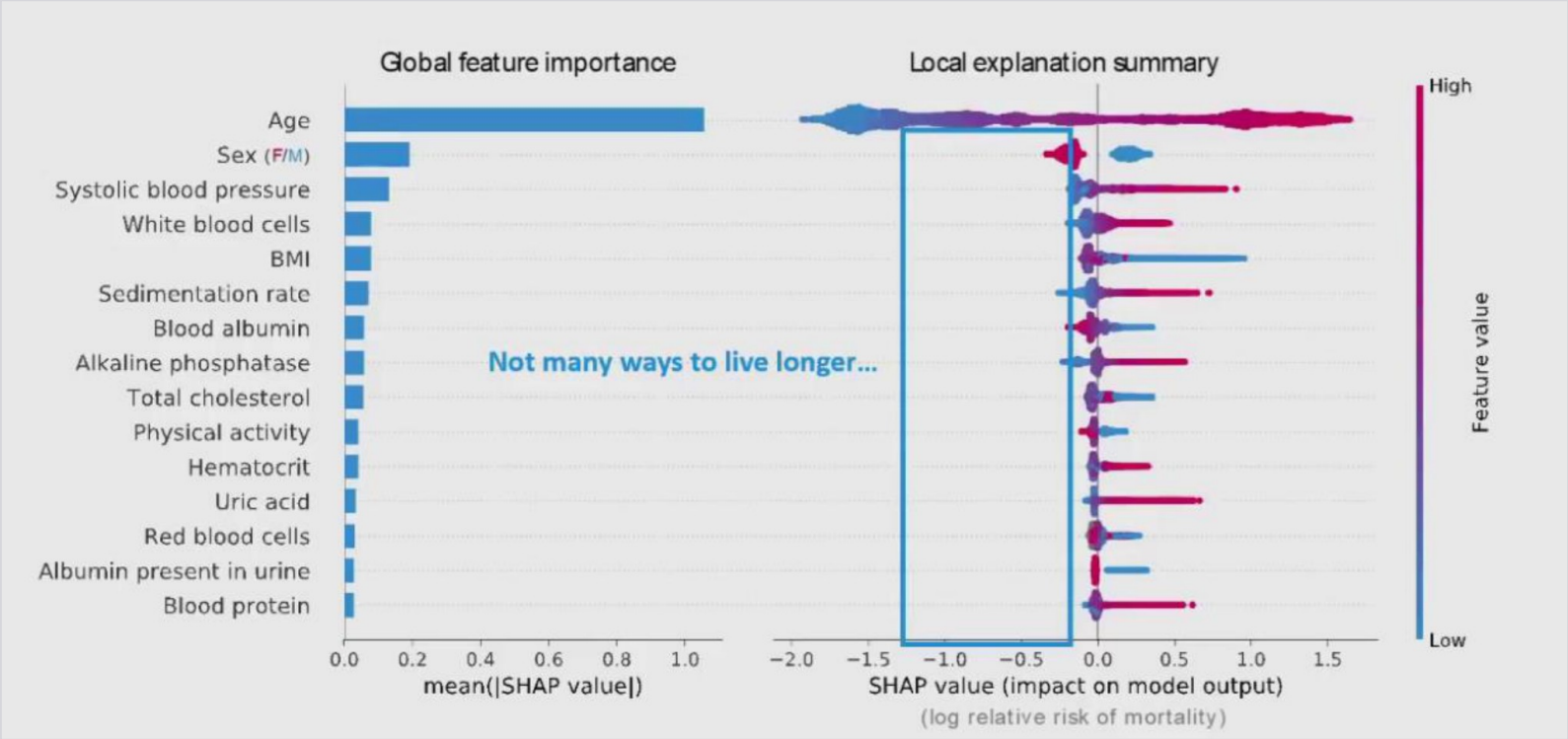


SHAP reveals individual effects hidden by global importance

Left: global feature importance (mean absolute SHAP value); right: per-individual SHAP value summary — log relative risk of mortality

How to read

- Feature contributions are computed for each individual (local)
- One dot is one person; color shows high or low feature values
- Box: few features pull toward living longer (left)

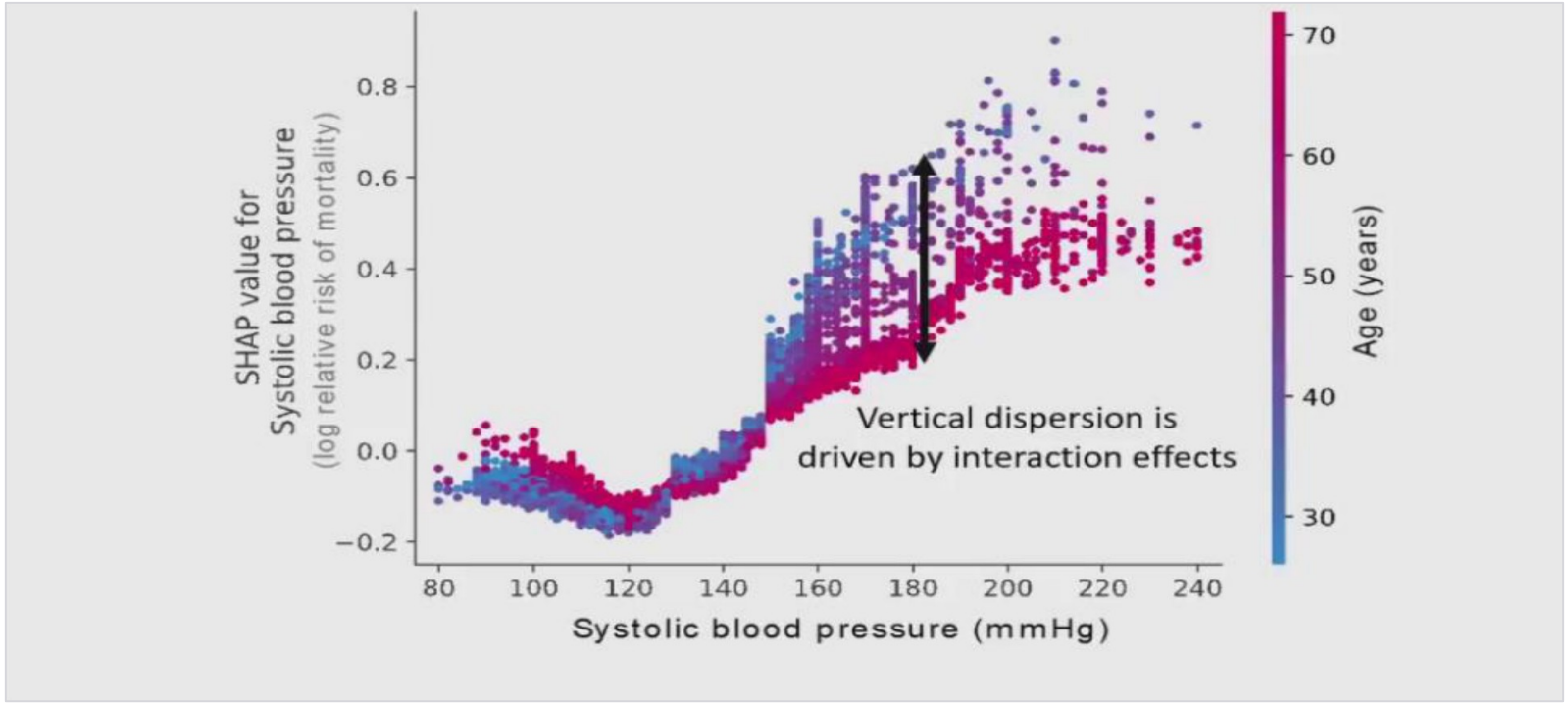


At the same blood pressure, risk contribution differs with age

Considering two features together — SHAP value of systolic blood pressure, colored by age

How to read

- Above 120, the risk contribution grows
- Vertical spread at the same blood pressure is the interaction effect
- Near 180, younger people (blue) have larger contributions

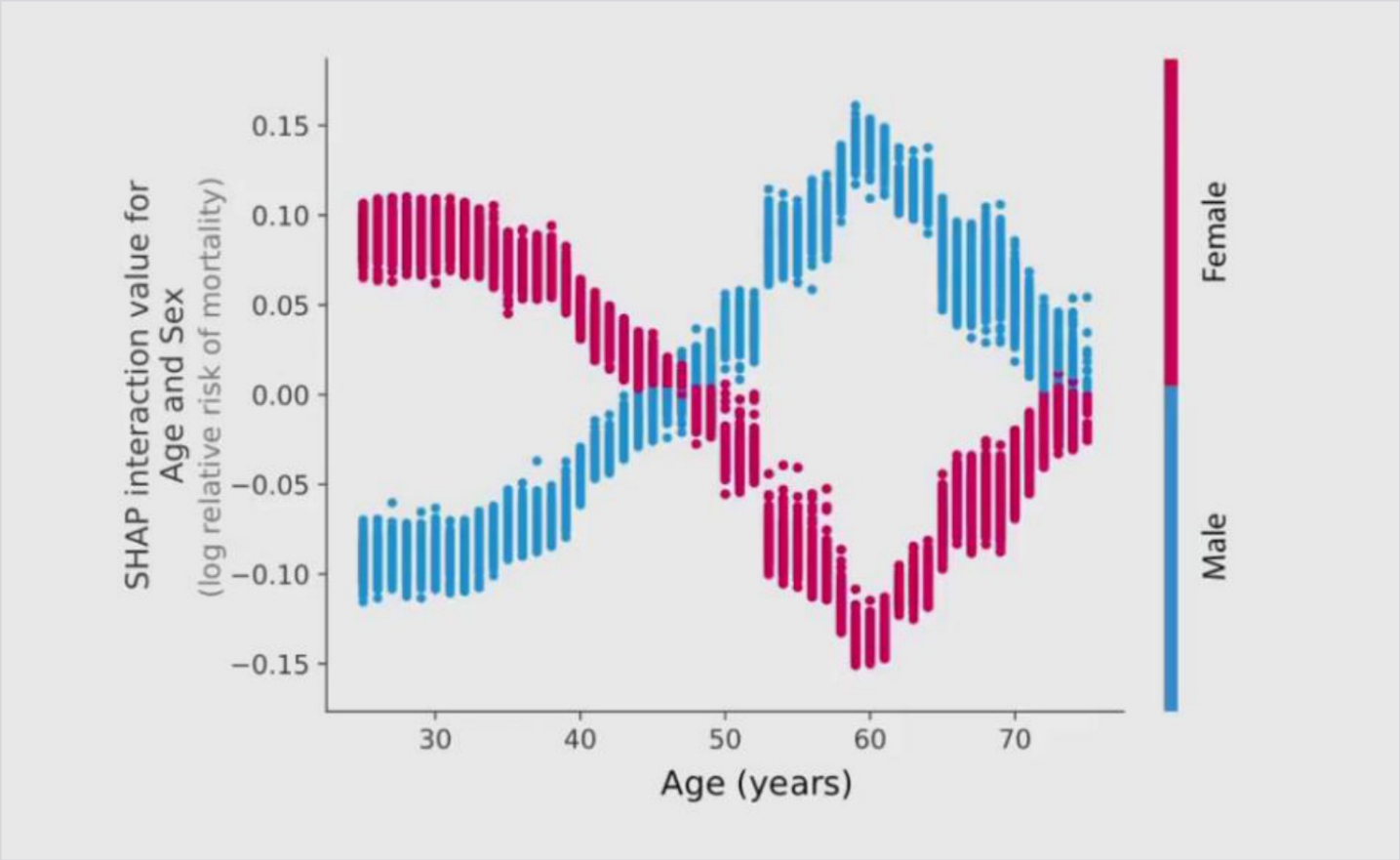


SHAP interaction values split the effect of age by sex

Decomposing one feature and considering it jointly — SHAP interaction value of age and sex

How to read

- Before age 40, the interaction value is positive for women (red), negative for men (blue)
- Around age 60 the direction flips, with men strongly positive
- One plot shows that the effect of age differs by sex

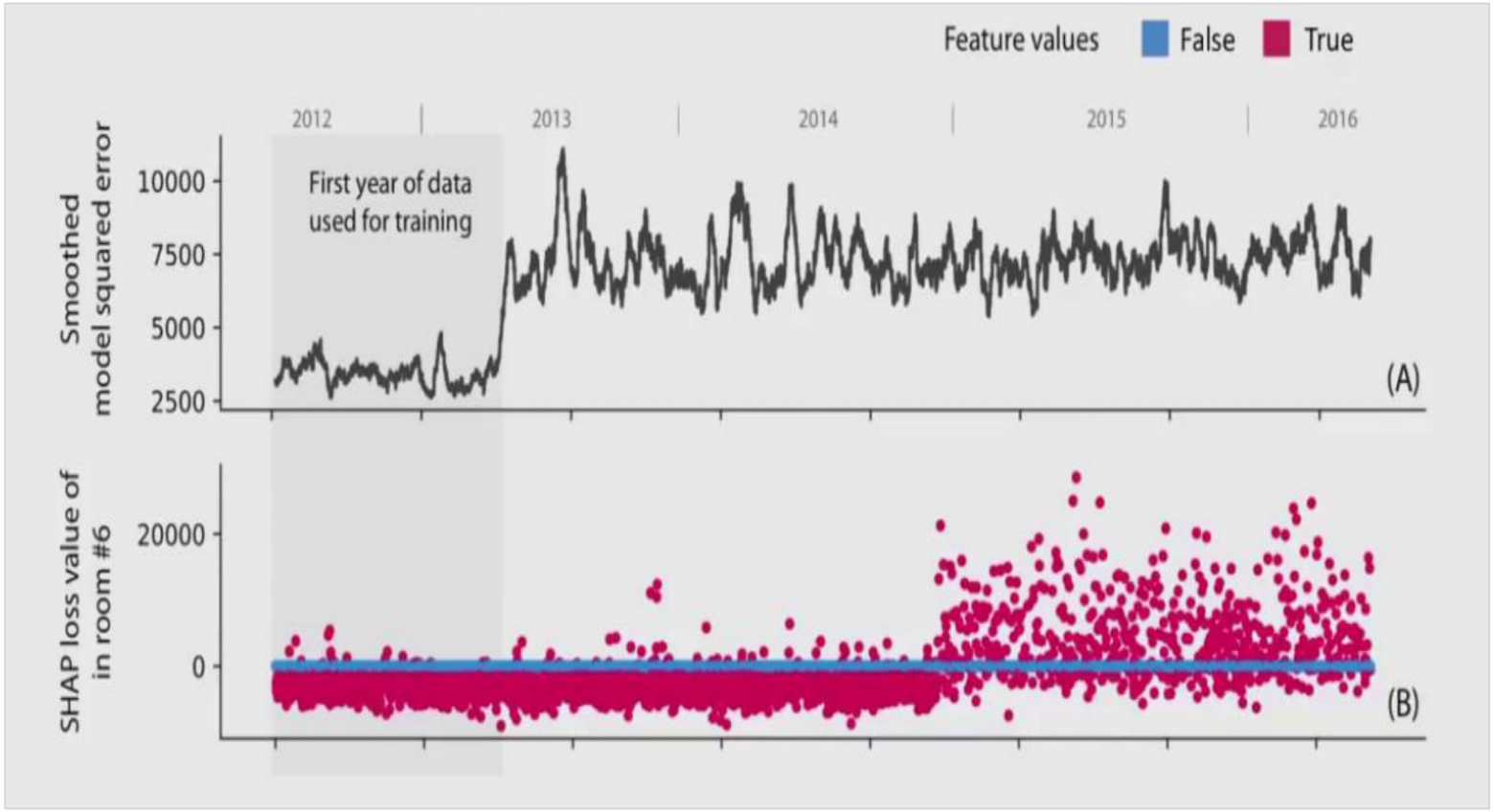


SHAP loss values trace the causes of post-deployment errors

Monitoring models over time — (A) smoothed model squared error, (B) SHAP loss value of the operating room 6 feature

How to read

- Only the gray period (first year) was used for training
- From around 2015, the loss contribution of the operating room 6 feature grows
- This pinpoints a specific feature as the cause of the rising error

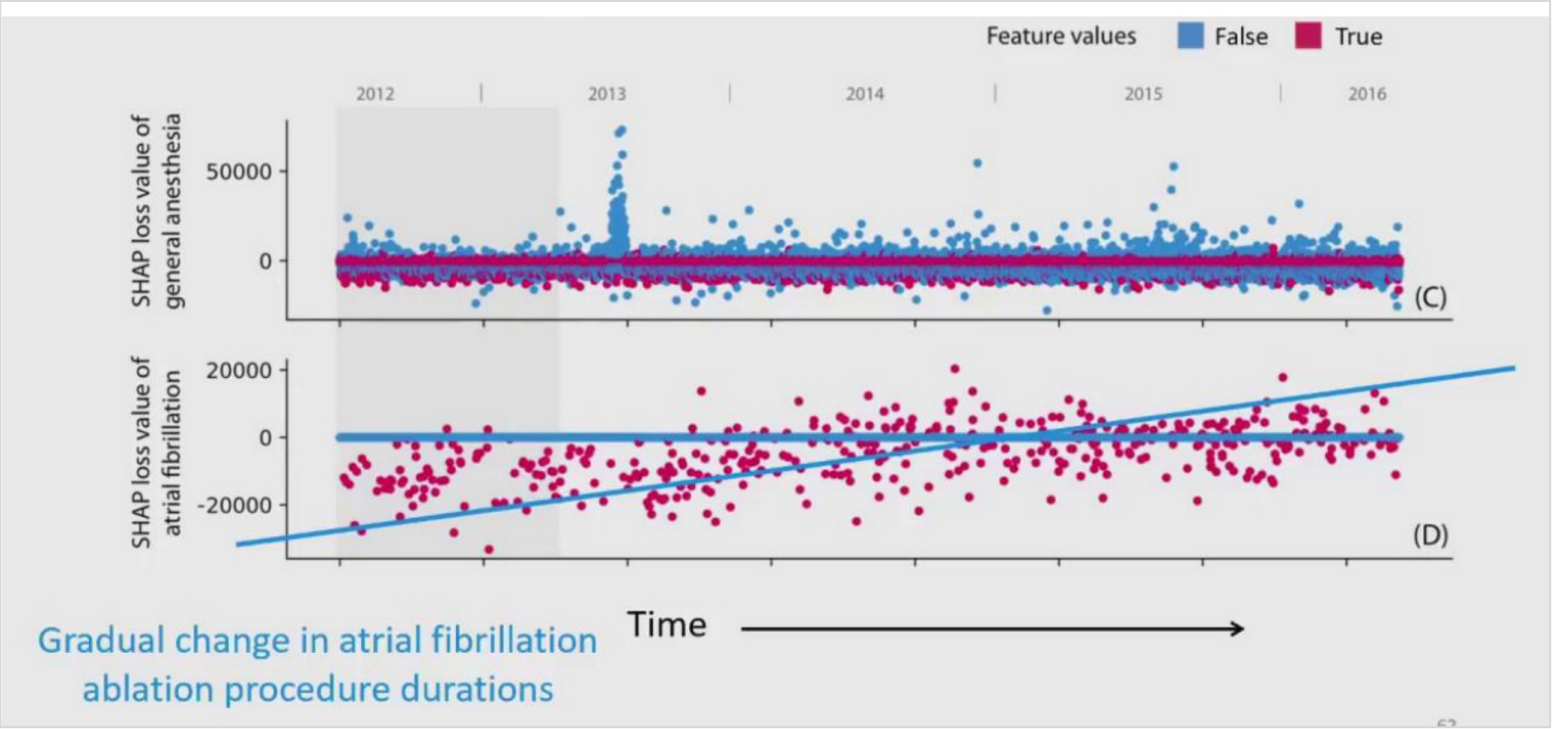


Gradual data change also appears as a SHAP loss value trend

Monitoring models over time — SHAP loss values of the (C) general anesthesia and (D) atrial fibrillation features

How to read

- (C) The general anesthesia feature shows a transient spike in early 2013
- (D) The loss contribution of the atrial fibrillation feature rises steadily over time
- This is because atrial fibrillation ablation procedure durations changed gradually



03

PART 3 · XGBoost Lab

Interpretable Machine Learning: Learning with XGBoost

Adult data — a model predicting whether annual income exceeds $50K

XGBoost's default importance ranks Age as the top feature

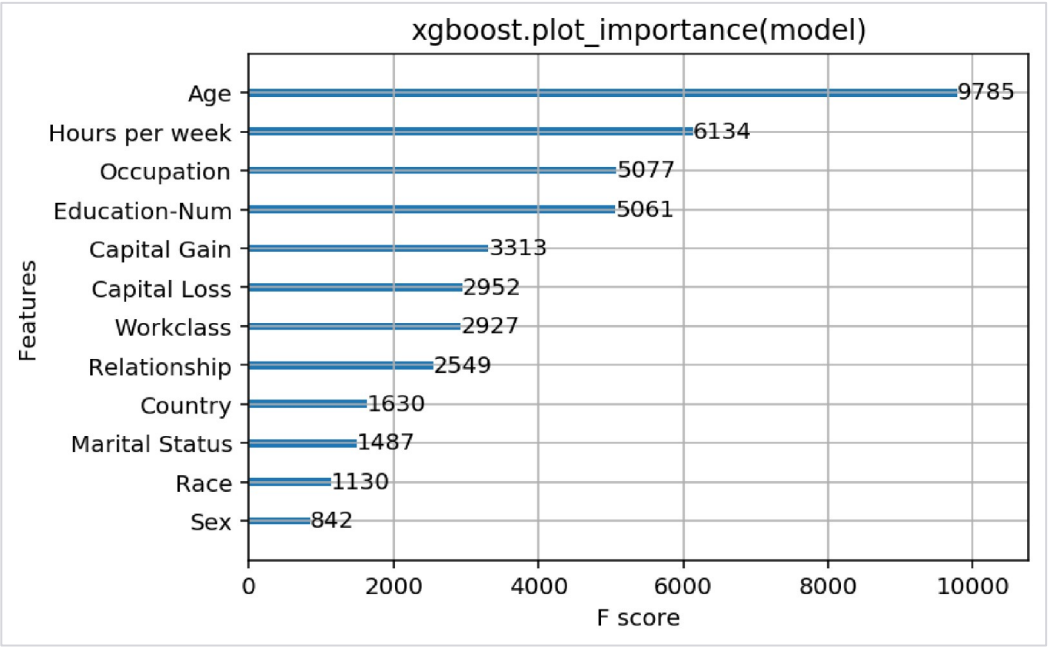
XGBoost's default importance plot — defaults to weight, the number of times used in splits

[Input] Prompt ① · Claude Code (Code tab)

```
In the adult-lab folder, please do the following.
1. Create a venv, install xgboost 3.4 and shap 0.54,
   and pin them in requirements.txt.
2. Load the data with shap.datasets.adult() and split
   80% train, 20% test (random_state=7).
3. Train a binary classifier: eta 0.01, max_depth 4,
   subsample 0.5, up to 5000 trees, early stopping 20.
4. Plot weight, cover, and gain importance as bar
   charts and save the ranks to importance.csv.
5. Use 2026 APIs and tell me about any warnings.
```

[Check] What a person verifies in the result

- Is #1 Age under weight, Capital Gain under cover, Relationship under gain?
- Did early stopping end training before the 5000-tree limit?
- Values may differ from the figure across versions; #1 should match



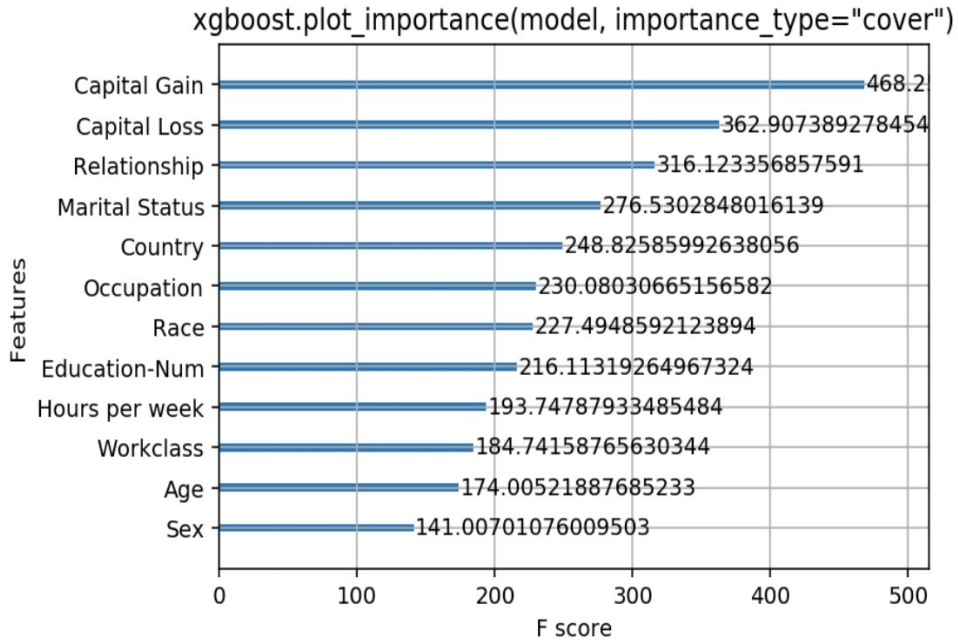
How to read

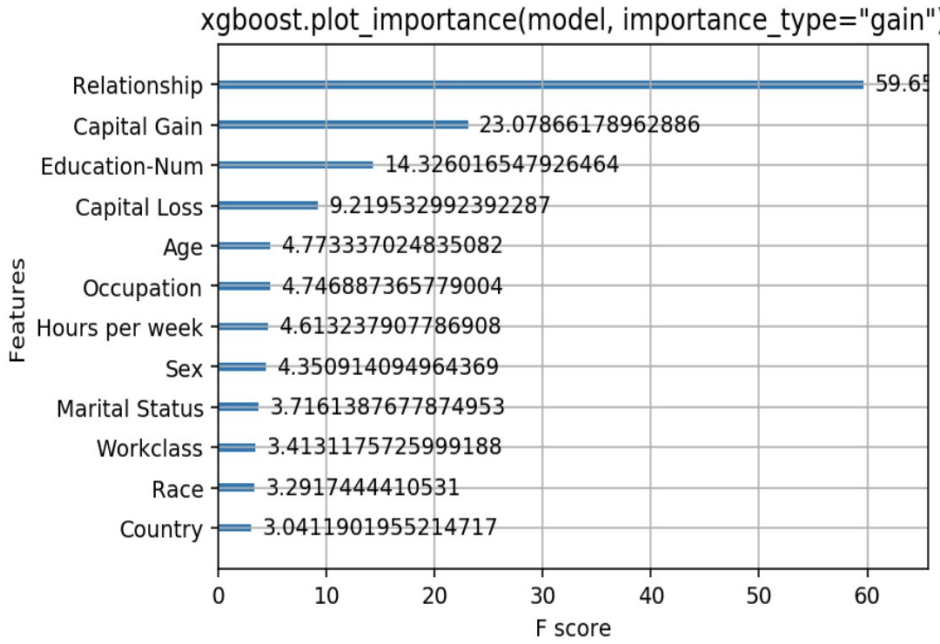
- The order is Age 9785, Hours per week 6134, Occupation 5077
- Is this really the order the model relies on?

Disappointingly, the feature order depends on the option

#1 is Age under weight, Capital Gain under cover (left), Relationship under gain (right)

- weight: number of times a feature is used to split data across all trees
- cover: number of times a feature is used in splits, weighted by the training data passing through them
- gain: average reduction in training loss from splits on the feature





SHAP values give importance that is consistent and additive

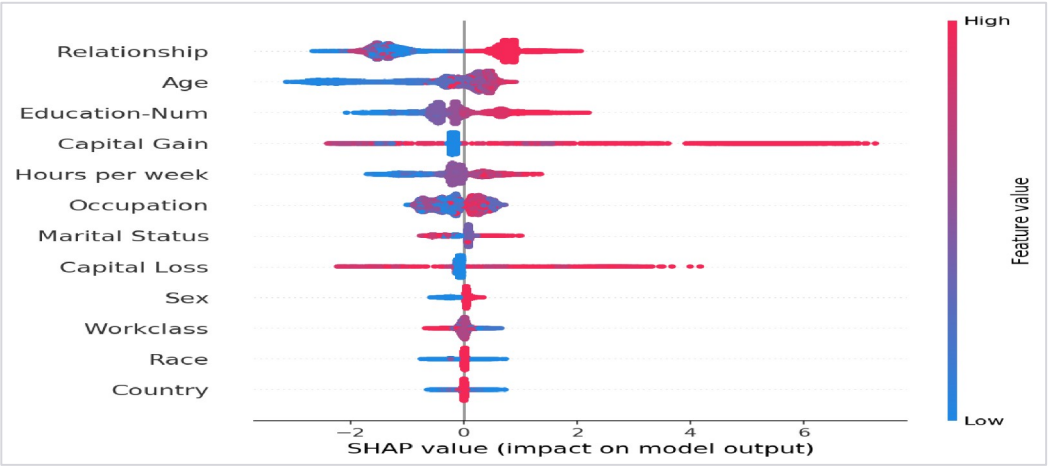
SHAP summary plot — one dot is one customer; color is the feature value (red high, blue low)

[Input] Prompt ② · Claude Code (Code tab)

```
Run a SHAP analysis on that model.
1. Compute SHAP values for all of X with TreeExplainer
   and confirm base value + SHAP = model output (log odds).
2. Save the summary plot as summary.png.
3. Save three dependence plots: Age, Age (color
   Education-Num), Hours per week (color Relationship).
4. Save the mean |SHAP| ranks to shap_rank.csv and
   explain the top 5 features' direction in one line each.
```

[Check] What a person verifies in the result

- Are the top three Relationship, Age, Education-Num?
- Do the base value and SHAP values sum to the model output?
- Is the Age plot negative in the early 20s, peaking at ages 40–60?



How to read

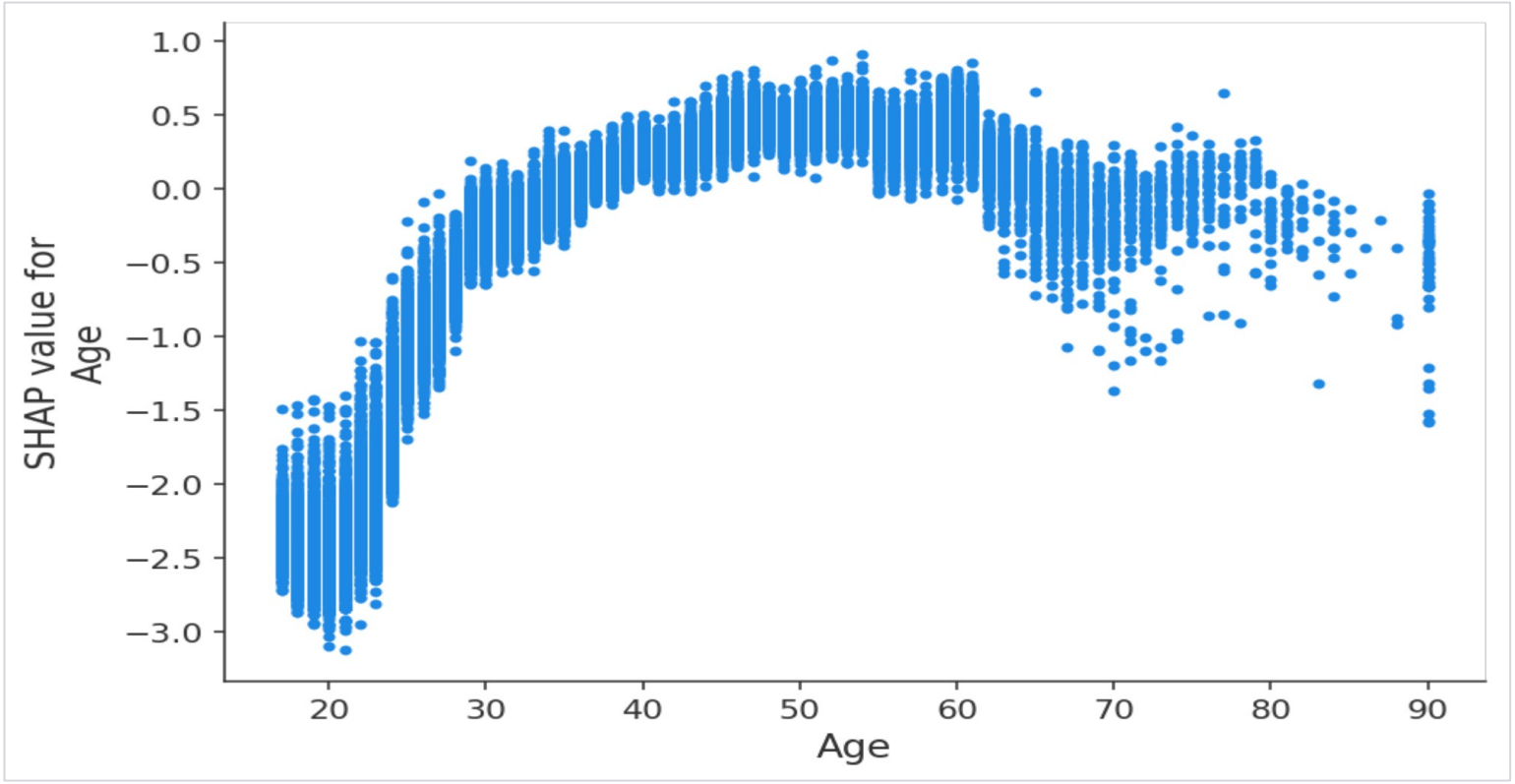
- Impact is largest in the order Relationship, Age, Education-Num
- Capital Gain is rare, but when large it raises the output sharply

Income contribution rises with age, then falls from the 60s

Features and SHAP values — SHAP dependence plot of Age

How to read

- y-axis: change in the log odds of earning over $50K a year as age changes
- Each dot is one customer in the dataset
- Strongly negative in the early 20s, peaking at ages 40–60

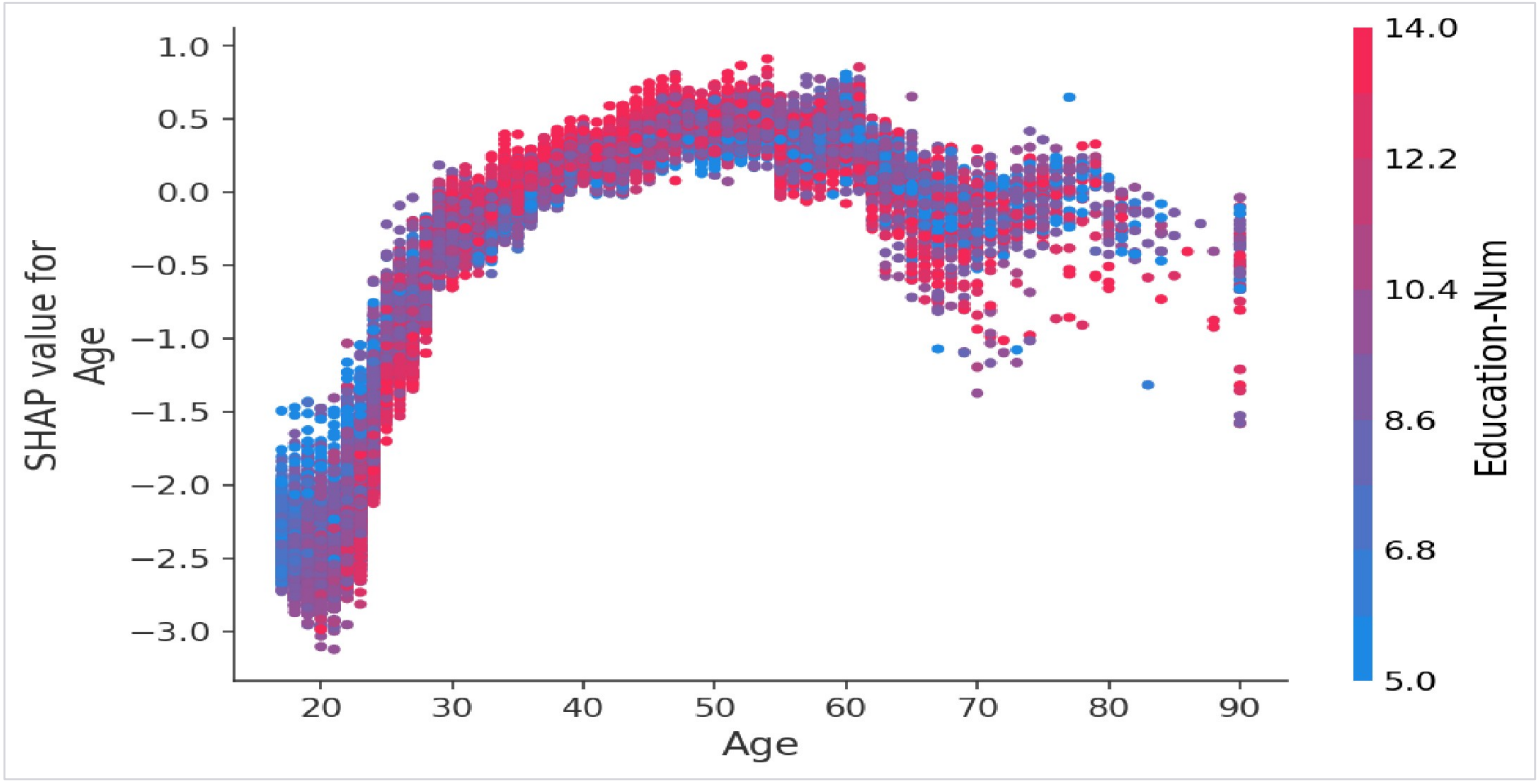


Adding color shows how age interacts with years of education

Features and SHAP values — SHAP value of Age, colored by Education-Num

How to read

- y-axis: change in the log odds with age
- Education-Num is the years of education a customer has completed
- Even at the same age, the contribution varies with years of education

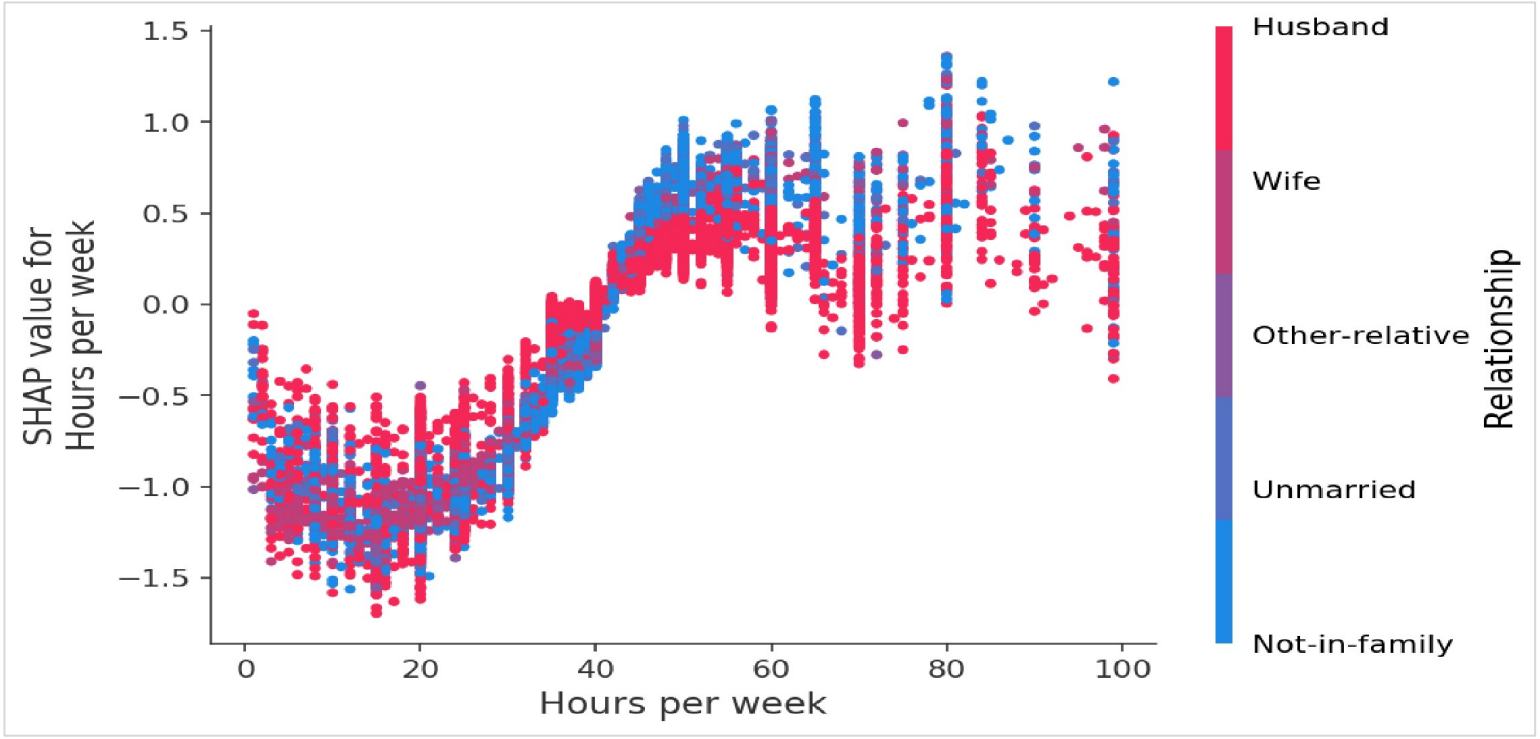


Beyond 40 hours a week, income contribution turns positive

Features and SHAP values — SHAP value of Hours per week, colored by Relationship

How to read

- Hours per week and their effect on earning potential
- Negative at 30 hours or fewer, turning positive at 40–50 hours
- Even at the same hours, the contribution varies by color (family relationship)



Explaining the evidence lets us trust and fix a model

Explanations are needed when we trust and use a model, fix it, and monitor it in operation

1

Why Explanations Are Needed

Screen out models that are right for the wrong reasons, and give the reasons for decisions such as loan rejections

2

Shapley Values and SHAP

Sum feature contributions per individual to rebuild a prediction, and read interactions and changes over time

3

XGBoost Lab

weight, cover, and gain disagree on the ranking. Compare with SHAP values, which are consistent and additive